INTRODUCTION

➢ Genetic diseases Overview

Genetic diseases are health conditions caused by abnormalities in an individual's DNA, such as mutations in specific genes or alterations in chromosome structure or number. These disorders may be inherited from one or both parents or may occur spontaneously due to new genetic mutations. Genetic diseases can affect various body systems and range from rare inherited disorders to common complex conditions.

Genetic disorders are broadly classified into three categories. **Single-gene (monogenic) disorders** are caused by mutations in a single gene, such as cystic fibrosis, sickle cell anemia, and Huntington's disease. **Chromosomal disorders** result from abnormalities in chromosome number or structure, for example Down syndrome and Turner syndrome. **Multifactorial disorders** arise from the combined influence of multiple genes and environmental factors and include conditions such as heart disease, diabetes, cancer, and Alzheimer's disease.

Genetic diseases occur due to inherited mutations passed from parents, spontaneous mutations that arise during the formation of reproductive cells or early embryonic development, or due to environmental triggers such as radiation, chemicals, and lifestyle factors. These genetic changes can disrupt normal biological functions and lead to disease manifestation at different stages of life.

Genetic diseases are medically significant as they contribute to a wide spectrum of human health conditions, from developmental disorders to life-threatening chronic illnesses. Understanding genetic factors is essential for accurate diagnosis, assessing familial risk, and enabling personalized medical interventions such as targeted therapies and preventive healthcare strategies.

➢ Challenges in predicting genetic diseases

Despite advances in medical science, predicting genetic diseases remains a significant challenge. Many common diseases are **multifactorial**, involving complex interactions between multiple genes and environmental influences. Additionally, variations in non-coding regions of DNA and the presence of genetic variants with uncertain significance make accurate interpretation difficult. The same genetic mutation may produce different symptoms in different individuals, a phenomenon known as variable expressivity, and in some cases, individuals carrying disease-causing mutations may never develop symptoms at all.

Another challenge in prediction is the delayed onset of many genetic disorders. Certain conditions manifest only in adulthood, making early identification difficult. Patients often experience prolonged diagnostic processes due to the complexity and rarity of genetic diseases, resulting in delayed treatment and increased psychological stress. Furthermore, genetic datasets may lack diversity or sufficient quality, affecting the reliability of predictive models.

➢ Importance of early risk prediction

Early risk prediction plays a crucial role in improving healthcare outcomes. Identifying individuals at higher genetic risk enables early intervention, preventive care, and informed clinical decision-making. Early prediction can reduce disease severity, improve quality of life, and lower long-term healthcare costs by shifting focus from treatment to prevention.

➢ Role of Machine Learning in Healthcare

Machine learning has emerged as a powerful tool in healthcare due to its ability to analyze large and complex datasets efficiently. Traditional diagnostic methods rely heavily on manual interpretation and struggle with high-dimensional genetic data. Machine learning techniques excel at identifying hidden patterns, correlations, and trends within large datasets, making them particularly suitable for genetic and biomedical applications. By leveraging machine learning, healthcare systems can improve predictive accuracy, reduce diagnostic time, and support personalized medicine.

➢ Brief Description of the Proposed Project

This project focuses on predicting the **risk of genetic diseases** using a family history-based dataset combined with genetic attributes such as age, gender, SNP values, and gene expression levels. The system employs machine learning techniques to analyse inherited patterns and genetic indicators to classify individuals as high-risk or low-risk. The proposed approach does not aim to diagnose specific diseases but rather to provide an early risk assessment to support preventive healthcare and informed medical decisions.

➢ Objectives

- To analyse family history and genetic attribute data
- To develop a machine learning-based model for genetic risk prediction
- To identify individuals with increased susceptibility to genetic diseases
- To support early awareness and preventive healthcare planning

LITERATURE STUDY

ML-based prediction of genetic diseases

➢ Title

Identifying disease genes using machine learning and gene functional similarities, assessed through Gene Ontology

Link:
https://pmc.ncbi.nlm.nih.gov/articles/PMC6287949/#:~:text=Supervised%20machine%20learning%20methods%20have,disease%20and%20non%2Ddisease%20genes.

➢ Authors and Year

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➢ Problem Statement

The problem statement of this is centered on the challenge of identifying disease-associated genes, particularly for complex diseases like Autism Spectrum Disorder (ASD), which exhibit highly heterogeneous genotypes. Despite advancements in genetic technologies and large-scale studies, identifying true disease genes remains difficult, as genetic etiology is only identified for 20-25% of ASD cases. The study aims to address this challenge by developing a machine learning-based methodology that leverages gene functional similarities assessed through Gene Ontology (GO) annotations to improve the identification and prediction of disease genes associated with complex diseases like ASD.

➢ Methodology

The study developed a machine learning-based approach to identify disease-associated genes, particularly for Autism Spectrum Disorder (ASD), using **Gene Ontology (GO)** annotations.

1. Data Preparation

- **ASD Genes**: High-confidence disease (HD) and low-confidence disease (LD) genes were obtained from the **SFARI gene database** as positive instances. Non-mental genes from the Krishnan et al. study were used as negative instances.

2. Functional Similarity Matrix Construction

- **GO Annotations**: Functional similarities between genes were calculated using GO annotations.
- **Semantic Similarity Measures**: Resnik, Wang, and Relevance (Rel) measures were used with the "Max combining strategy" to compute similarity scores.

3. Machine Learning Classifiers

1. **Random Forest (RF)**: Best-performing classifier with 500 trees and equal class weights.
2. **Naive Bayes (NB)**: Probabilistic classifier based on Bayes' theorem.
3. **Linear-SVM**: Linear Support Vector Machine.
4. **Radial-SVM**: Non-linear SVM with a radial kernel.

4. Classifier Evaluation

1. **Stratified Five-Fold Cross-Validation**: Classifiers were trained on four folds and tested on the fifth.
2. **Held-Out Restricted Validation**: Classifiers were trained on HD + LD genes and non-mental genes but tested only on HD genes.

5. Addressing Class Imbalance

- **Undersampling**: Balanced the dataset with 30% positive (disease genes) and 70% negative (non-disease genes).

6. Automated Workflow

- Developed a **KNIME workflow** to automate the methodology, allowing researchers to test classifiers and functional similarity measures without programming expertise.

7. Prediction of Novel ASD Genes

- The best-performing RF classifier (trained on Resnik scores) predicted 73 novel ASD candidate genes from 554 genes disrupted by rare de novo CNVs in ASD patients. These genes were analyzed for functional enrichment using the **Enrichr tool**

➢ Dataset Used

The study utilized the following datasets to train and evaluate the machine learning models:

1. **ASD Genes**:
    - Genes associated with Autism Spectrum Disorder (ASD) were obtained from the **Simons Foundation Autism Research Initiative (SFARI) gene database** (N = 990), accessed in March 2018.
    - SFARI genes are categorized into seven categories based on the strength of evidence for their association with ASD:

        - **High Confidence Disease (HD) Genes**: Genes with strong and reproducible evidence
        - **Low Confidence Disease (LD) Genes**: Genes with minimal evidence
        - Syndromic genes and genes with hypothesis-based evidence or unscored genes were excluded .
    - HD and LD genes were used as positive instances (disease genes).
2. **Non-Mental Genes**:
    - Non-mental genes (N = 1189) were obtained from the Krishnan et al. study. These genes were used as negative instances (non-disease genes).
    - 49 genes from the non-mental gene dataset were found in the SFARI database and reassigned as positive instances.

Model Training:

1. **Functional Similarity Matrix Construction**:
    - Functional similarities between genes were calculated using three semantic similarity measures: **Resnik**, **Wang**, and **Relevance (Rel)**.
    - The "Max combining strategy" was used to compute the maximum semantic similarity score between pairs of GO terms associated with genes.
2. **Machine Learning Algorithms**:
    - Four supervised machine learning classifiers were used: **Random Forest (RF)**, **Naive Bayes (NB)**, **Linear-SVM**, and **Radial-SVM** and trained on functional similarity matrices of disease (ASD) and non-disease (non-mental) genes.
3. **Evaluation**:
    - Two validation strategies were employed:
        - **Stratified Five-Fold Cross-Validation**
        - **Held-Out Restricted Stratified Five-Fold Cross-Validation**
4. **Class Imbalance Handling**:
    - Undersampling was applied to address class imbalance, with the majority class (non-disease genes) reduced to 70% of the dataset, while the minority class (disease genes) was set to 30%.
5. **Classifier Performance**:
    - The performance of the classifiers was evaluated using the **Area Under the Receiver Operating Characteristic Curve (AUC)** metric.

➢ Key Results

- The best-performing Random Forest classifier (trained on Resnik semantic similarity scores) was used to predict novel ASD candidate genes from a set of 554 genes disrupted by rare de novo Copy Number Variants (CNVs) in ASD patients.
- The classifier predicted 73 novel ASD candidate genes, which were analyzed for functional enrichment using the **Enrichr tool**.
- The Random Forest classifier trained on Resnik semantic similarity scores achieved the best performance (AUC = 0.80 for HD genes and 0.82 for HD + LD genes in held-out validation).
- The study demonstrated that machine learning methods trained on GO-based gene functional similarities can outperform existing methods in predicting disease genes

REQUIREMENT ANALYSIS

➢ Problem Definition

What real world problem exists ?
Genetic diseases are complex and difficult to identify due to the intricate nature of genes and DNA, which are not directly observable. It is often unclear whether a disease occurring in an individual is hereditary or acquired, making early prediction challenging.

Why genetic diseases are hard to predict?

Genetic disorders may arise due to inherited gene mutations, spontaneous mutations during embryonic development, or may skip generations and appear later with varying symptoms. Family medical history provides valuable insight into inherited gene patterns, mutation trends, and underlying risk factors.
The objective of this system is to analyze family history and genetic attributes to predict an individual's risk of developing genetic diseases at an early stage.

➢ Existing System

How genetic diseases are currently identified ?
Current methods for identifying genetic diseases include techniques such as Next-Generation Sequencing (NGS), Chromosomal Microarray Analysis (CMA), Molecular Testing (PCR), Cytogenetic Analysis, Biochemical Testing, and Non-Invasive Prenatal Testing (NIPT). These methods primarily focus on diagnosing genetic disorders after symptoms appear.

Late detection and limitations
Late detection often results in missed opportunities for early intervention, irreversible disease progression, psychological distress, and prolonged diagnostic processes. Additionally, existing systems may struggle to accurately predict disease severity or onset and may produce false positive or false negative results. Many tests provide incomplete information and do not fully account for inherited risk patterns.

➢ Proposed System

What this system will do differently ?
The proposed system utilizes machine learning techniques to analyze family history and genetic attributes such as age, gender, SNP values, and gene expression levels to predict an individual's risk of developing genetic diseases.

Rather than diagnosing specific diseases, the system focuses on identifying individuals with increased genetic susceptibility.
This risk-based prediction approach supports early awareness, preventive care, and informed medical decision-making while reducing dependence on late-stage diagnostic procedures.

➢ Stakeholders

- **Patients:** Benefit from early risk assessment and preventive care
- **Hospitals and Clinics:** Use the system as a decision-support tool
- **Medical Researchers:** Analyse genetic risk patterns and model accuracy
- **System Developers:** Responsible for system maintenance and updates

➢ Functional Requirements

- The system shall collect and process family history and genetic data accurately
- The system shall pre-process and clean the dataset before analysis
- The system shall train machine learning models to predict genetic risk
- The system shall classify individuals as high-risk or low-risk
- The system shall minimize false negative predictions to ensure patient safety

➢ Non-Functional Requirements

- **Performance:** The system should generate predictions within milliseconds
- **Accuracy:** The prediction accuracy should exceed 95% with minimal false negatives
- **Security:** Genetic and family data must be protected from unauthorized access
- **Scalability:** The system should handle increasing dataset sizes efficiently
- **Usability:** The system is intended for use by trained professionals rather than general users

➢ Data Requirements

Type of dataset and its attributes

The dataset used is a static dataset stored in an Excel format.
The dataset includes attributes such as age, gender, family history indicator, SNP values (1–5), gene expression values (1–5), and a result label. Family history data is crucial as genetic traits and mutations are often inherited, increasing the likelihood of disease risk across generations.

➢ Software Requirements

- Programming Language: Python
- IDE/ Development Tools: Spyder , PyCharm , Sublime Text
- API testing tool : Postman
- Database Services : Firebase
- Frameworks/Libraries: Pandas, NumPy, Scikit-learn, Flask
- Interface Tools: Tkinter

➢ Hardware Requirements

- Minimum 4 GB RAM
- Standard processor (Intel i3 or equivalent)
- Stable internet connection

➢ Constraints

- Genetic attributes such as SNP and gene expression values are not easily obtainable through surveys
- Data privacy and ethical concerns restrict dataset accessibility
- System accuracy is highly dependent on dataset quality and reliability